

Research Questions and Objectives

Level 1 - Research Foundations | Student Learning Pack

How to use this pack

Read the teaching sections first, complete the worked example, do the short practice task, then take the basic knowledge check in RSRE. External references are optional further learning; you should be able to answer the core assessment from the Academy material.

Lesson 1: What makes a question researchable?

A research question identifies what you want to find out. A strong question is clear, focused, feasible and answerable with evidence. It identifies a population, issue, exposure, experience, intervention or outcome where appropriate.

Compare: “Why is diabetes a problem?” is too broad. “What proportion of adult patients attending X clinic have controlled blood pressure?” is more specific. The best wording depends on the design and purpose of the study.

Lesson 2: Questions, objectives and hypotheses

The question states what you want to know. The objective states what the study will do. A hypothesis is a testable expectation used when appropriate. For example: Question — “Is sleep duration associated with academic performance among medical students?” Objective — “To assess the association between sleep duration and academic performance among medical students at X university.”

Objectives should be specific enough to guide the methods and analysis.

Lesson 3: Build one step at a time

Start with a health problem. Write one sentence describing the problem, one research question, one primary objective and two secondary objectives. Remove words that do not help define what will actually be measured or explored. Then ask: Could another researcher tell what evidence I need from this wording?

Lesson 4: Worked example

Problem: students often report difficulty accessing research articles. Question: “What barriers do health-science students report when accessing scholarly articles?” Objective: “To describe perceived barriers to accessing scholarly articles among health-science students at X institution.” Secondary objectives could explore differences by year of study or access to institutional subscriptions.

Notice that the wording points directly toward data collection.

Lesson 5: Check your understanding

Before the quiz, remember the sequence: problem → question → objective → method. Do not choose a statistical test or questionnaire first. Start by deciding exactly what you need to learn.

Competency check

After completing this module, you should be able to explain the core idea in your own words, identify a correct and incorrect application, and make a basic research decision using the concept.

RSRE note: The Academy is optional. Experienced researchers may use the diagnostic/competency pathway and enter other RSRE pillars directly.